

The Axon Tract as a Transducer: Simulation of Gravitational Waves Coupling With an Ion at the Node of Ranvier

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Abstract

What happens when an axon tract acts as an intermediary between metric perturbations and ions in a node of Ranvier of an axon in the tract? In this paper, the authors present computer simulations of the time dependent Schrodinger Equation that utilize Crank-Nicolson discretization. The environment potential is provided by action potentials of two neighboring axons, which in turn are perturbed by gravitational waves. At about fourteen significant figures in MATLAB precision, a clear impact of environmental potential modification on the ion wave function evolution is verified. Betwixt the metric perturbation and the quantum evolution, the axon tract can thus be seen as a classical-quantum transducer.

1 Introduction

Weinberg (Weinberg, 2012) and several other authors have considered the role of gravity in the collapse of the quantum wavefunction. The present paper does not consider this important question directly, rather, here we wish to explore further the mechanism of universal quantum computation in the brain proposed in (Chawla, 2022) and (Chawla & Morgera, 2022b). Towards this end, we set up a MATLAB simulation of the time dependent Schrodinger equation for a single massive particle (representing, say, a sodium ion) at a node of Ranvier. Our motivation is to couple this simulation with the simulation which provides minute timing-differentials in axon tract action potentials when a gravitational wave is impinging on the tract versus when it is not. We are successful in our objective and, when we use MATLAB's high precision, we are able to see an impact on the wave function's evolution - an impact dependent on the environmental potential. Such biophysical quantum simulations have not been performed previously in the literature, to the best of our knowledge. The significance of these simulations is that they form a stepping stone towards a similar study that looks at decoherence due to ion-ion collisions at the node of Ranvier, a task which we will take up in forthcoming work.

The paper is organized as follows. In Section 2 we set up the simulation formulae for the discretized Schrodinger Equation for a single particle, where the standard Crank Nicolson method is used. The result of coupling this simulation with the programs developed in (Chawla *et al.*, 2019) and (Chawla *et al.*, 2021) is presented in Section 3. Finally, we conclude in Section 4.

2 Crank Nicolson Approach to the TDSE

Consider the standard time-dependent Schrodinger equation (TDSE) (Griffiths & Schroeter, 2018),

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi \quad (1)$$

where i is the complex number $\sqrt{-1}$, $\hbar$ is the Planck constant divided by 2π , $\psi(x, t)$ is the wave function of interest, dependent on x and t , m is the mass of the particle and V is the “environment” potential in which the particle is placed. To simulate this equation, we will do a discretization first. Thus, we will write $\psi(x, t)$ as $\psi(i, j)$ where i is the spatial index and j is the temporal index. The first partial derivative with respect to time can be expressed in the form

$$\frac{\partial \psi}{\partial t} \rightarrow \frac{(\psi(i, j+1) - \psi(i, j))}{\Delta t}. \quad (2)$$

Similarly,

$$\frac{\partial \psi}{\partial x} \rightarrow \frac{(\psi(i+1, j) - \psi(i, j))}{\Delta x}. \quad (3)$$

And thus, the second partial derivative with respect to space can be written as

$$\frac{\partial^2 \psi}{\partial x^2} \rightarrow \frac{(\psi(i+1, j) - 2\psi(i, j) + \psi(i-1, j))}{(\Delta x)^2}. \quad (4)$$

Plugging these discrete versions into the primary Schrodinger equation, we get

$$i\hbar \left[\frac{(\psi(i, j+1) - \psi(i, j))}{\Delta t} \right] = -\frac{\hbar^2}{2m} \left[\frac{(\psi(i+1, j) - 2\psi(i, j) + \psi(i-1, j))}{(\Delta x)^2} \right] + V(i, j)\psi(i, j). \quad (5)$$

Next, we will keep the $(j+1)$ term on the LHS and move all other (j) terms to the RHS. We get,

$$i\hbar \left[\frac{\psi(i, j+1)}{\Delta t} \right] = i\hbar \left[\frac{\psi(i, j)}{\Delta t} \right] + \frac{\hbar^2}{m} \left[\frac{\psi(i, j)}{(\Delta x)^2} \right] + V(i, j)\psi(i, j) - \frac{\hbar^2}{2m} \left[\frac{\psi(i+1, j)}{(\Delta x)^2} \right] - \frac{\hbar^2}{2m} \left[\frac{\psi(i-1, j)}{(\Delta x)^2} \right]. \quad (6)$$

This yields,

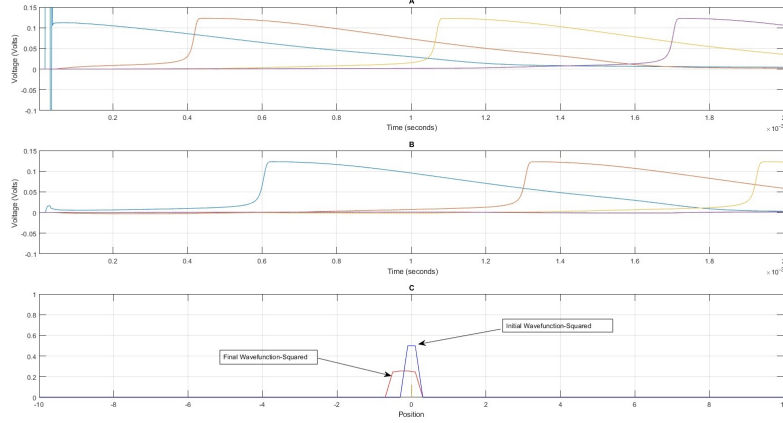


Figure 1: Shows (A) the first coupled axon which is stimulated, (B) the second coupled axon which responds after a delay and (C) the initial and final wave functions (absolute-squared) that are the outcome of evolution under a potential formed from (A) and (B) and shown in the next figure. In (A) and (B) the x-axes are time in seconds and the y-axes are voltage in volts. In (C), the x-axis is position and the y-axis is the squared magnitude of the wavefunction.

$$i\hbar\left[\frac{\psi(i,j+1)}{\Delta t}\right] = \left[\frac{i\hbar}{\Delta t}\right] + \left[\frac{\hbar^2}{m} \frac{1}{(\Delta x)^2} + V(i,j)\right]\psi(i,j) - \frac{\hbar^2}{2m}\left[\frac{\psi(i+1,j)}{(\Delta x)^2}\right] - \frac{\hbar^2}{2m}\left[\frac{\psi(i-1,j)}{(\Delta x)^2}\right] \quad (7)$$

The above can be easily put into matrix form and thereafter put into MATLAB code form.

3 Simulation Results

We take the above model and use the MATLAB program developed in the gravitational wave section of (Chawla *et al.*, 2021) to provide us the potential. We vary the potential and look for an impact on the TDSE-governed evolution. When we utilize upto 14 significant figures in MATLAB, we are able to verify that there is indeed a link between the potential and the evolution; varying the potential changes the evolved state. This is consistent with (Chawla, 2022; Chawla & Morgera, 2022b,a).

The results of simulation are presented in Figure 1, subplot 1, which shows us the first axon's potential train while subplot 2 shows the second (unstimulated) axon's potential train. The final subplot of Figure 1 demonstrates evolution in the wavefunction under the evolving environmental potential shown in Figure 2.

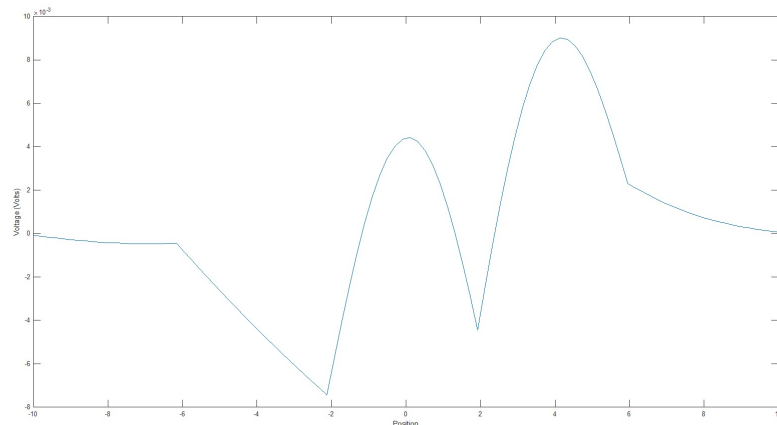


Figure 2: Shows the time-varying potential which is a difference of the two axons' instantaneous voltage profiles, taken at the last step of the evolution. The x-axis is position and the y-axis is voltage in volts.

4 Conclusion

It was reported in (Chawla & Morgera, 2022a) that high precision MATLAB simulations yield an interaction between metric perturbations and the differential response timing of coupled axons in an axon tract. As the capabilities of neural recording devices are improved, novel ideas such as quantum sensing (Hansen *et al.*, 2022) can be incorporated to measure these minute effects. The next step, interventions in this mechanism, can be carried out, with the aim of studying the impact on subjective perception, especially spacetime perception (for example, (Burr & Morrone, 2006)).

The paper has certain limitations in that the various constants such as Planck's constant and the mass of the simulated particle, were not insisted on being accurate. This can be rectified in future work. In forthcoming work, we will extend this study to the decoherence that ensues out of collision of ions at the node. A further direction of investigation is quantum noise and how it can be studied in relation to classical noise at the ion channel level in the context of the present work. Overall, this paper offers another step forward in elucidating the mechanism of quantum computing in the brain. We must note here that this also has implications for significant questions such as the existence of free will (Lloyd, 2012).

References

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